

Importance of Change Management in Digital Transformation Sustainability

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Abstract: Organizational changes triggered by digital transformation usually focus only on internal processes or yet baser on technology, leaving behind, or even totally ignoring the consequences of such changes on people. There is a lack of studies presenting the pathways for sustainable change management during digital transformation, especially those from traditional professions, less digitally mature businesses, and developing countries (Philip J., 2021). Digital Transformation plays an important role in improving the efficiency of company's operations which links to sustainability. The business transformation will be successful only when employees are wholeheartedly committed to the company's vision and not merely when the company successfully installs some of the new and emerging technologies and implements changes (Florek-Paszkowska, 2021). Yet, the ultimate studies show that digital transformation initiatives have still very high tendency to failure. Aiming to resolve this issue, this study conducted a systematic literature analysis to review articles that discuss current trends in the Change Management and Digital Transformation discipline.

The study results propose a Change Management Framework composed of its three imperatives: Leadership, Communication and Inclusion, which on the other hand induce strong foundations of a sustainable and resilient digital transformation in organizations. This paper synthesizes the presented research field's importance and provides insights for practitioners how to have their digital transformation initiatives succeed and retain sustainable.

Originality/value – The study analyzes the impact of change management to digital transformation sustainability through a holistic approach, conventionally not given enough attention even though it may be found in every single research as a critical factor to digital transformation success.

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Keywords: Digital Transformation, Change Management, Resistance, Technology, Digitalization, Digital innovation, Sustainability, Challenges, Barriers, Success factors

1. INTRODUCTION

“Digital Transformation” is not a novel concept, but as it suddenly, due to covid-19 pandemics, swapped its role from a ‘luxury’ initiative of a modern organization, into the ‘necessity’ for the organization to survive, the notion has become so commonly heard in everyone’s daily life as we could freely attribute it the epithet of the biggest trend of 2020. Digitalization is radically drifting the business landscape across industries and to keep competitive advantage over the market, businesses around the world are increasingly investing in digital transformation. According to the International Data Corporation (IDC), global spending on the digital transformation (DX) of business practices, products, and organizations is forecast to reach \$2.8 trillion in 2025, more than double the amount allocated in 2020. This forecast is fairly attributed to the fact that entrepreneurs at the large scale are becoming aware of the digital transformation being the core driver of their organization’s sustainable competitive advantage.

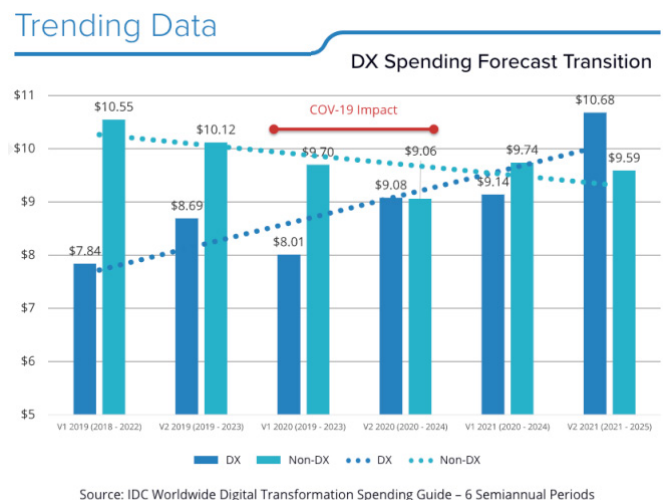


Fig. 1. Spending Forecast in Digital Transformation (IDC, 2021)

Digitization is denoted as a tool to accelerate a sustainability transition. Cloud computing has enabled easier access to

services by reducing costs of digital infrastructure. Artificial Intelligence/Machine Learning (AI/ML) is bringing new capabilities to various industries, which could add 14% to global GDP by 2030 (PwC, 2017). Powered by 5G, the Internet of Things (IOT) will connect billions of devices, making everything in homes, offices, and factories smarter. It is the last call to both practitioners and researchers to give it a thorough attention not only the digital economies but also their sustainability and resilience. Although digital transformation plays an important role in promoting the sustainable competitive advantage of the enterprise, there are still many problems and far too many challenges accompanying the implementation of such projects.

There is lots of research found in literature regarding the success factors and its opponents, the barriers and challenges in information system and DT implementations; and while the change management element can be found, literally, in each of these studies, it still lacks the adoption of a holistic approach for better understanding the impact of change management to digital transformation sustainability. *Starting a digital transformation journey without a dedicated focus on Change Management is like leaving the safe harbor with no knowledge of how to sail through turbulent waters (Siemens, 2019).* To adequately address this issue, this study will use the systematic literature review with the aim to investigate the impact of change management to a resilient and sustainable digital transformation in a holistic manner.

2. METHODOLOGY

Depending on the research objectives, one could consider different review types, such as: narrative review, descriptive review and theoretical review. For the Digital Transformation and Change Management discipline, narrative reviews and theoretical reviews are the most common review studies (Paré et al., 2015). Given that IT and especially change management schemes are social-oriented technologies, digital transformation and change management require detailed systematic literature reviews, as these can point out relevant studies by using explicit methodology (Paré et al., 2015; Khan et al., 2003).

This study will use systematic literature review aiming to examine and highlight the importance of change management in the sustainable digital transformation journey.

Data collection was done by using a keyword search. Different terms were used for Change Management (such as: organizational change, resistance, barriers to successful DT, challenges), Digital Transformation (such as: digitalization, business transformation, Information systems, business process reengineering) and Sustainability (in organizations, IS sustainability, green technology). And these terms were searched for in titles, abstracts, keywords, and keyword plus. More importantly, this study used mostly recent studies from which more than ten papers are from the last three years in order to assess all the fast paced developments in Digital Transformation discipline that happened as the result of the recent pandemics with Covid-19.

3. LITERATURE REVIEW

A fourth annual digital transformation survey conducted in January 2022 by Harvard Business Review Analytic Services of 727 executives from around the globe finds that 92% say their organization's digital transformation will become more important to business success over the next 12 months (Fig. 1).

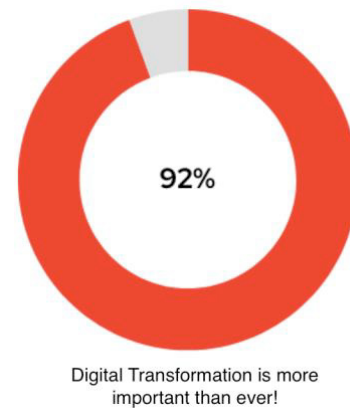


Fig. 2. Importance of DT to business success (HBR, 2022).

Digital transformation is rarely easy to succeed, but effective change management makes change feel less difficult and brings the benefits of change into sharper focus, both of which make employees more likely to support change and help to make it stick (APQC, 2022). Change Management is a top three concern among executives when entering digital transformation projects (Siemens, 2019). Also in academia, numerous authors stress the importance of change management as it helps organisations to address problems such as resistance to change by encouraging users to understand and own the system (Odhiambo, 2010; Singh & Wesson, 2009; Smuts, Van der Merwe, Loock, & Kotze, 2010).

Change management is a collective term for all approaches to prepare and support individuals, teams, and organizations in dealing with the transition or transformation of an organization's goals, processes or technologies, in literature denoted as organizational change. Change management can be referred to as the capability to anticipate change which may happen in the future in order to effectively manage it (Kalema, Olugbara, & Kekwaletswe, 2014). Changing the ways of thinking or perceiving often causes resistance due to the fact that the established way is based on what we have learned in a group that we belong to. The need to be accepted by and belong to a group is unconscious and therefore powerful and even if we privately disagree, we don't want to deviate from that group behaviour because this might make us lose group membership or may violate some aspect of our identity (Schein, 2004, p.321).

The analysis shows that prior studies identified people as both the biggest challenge and main source of success to DT. Toytari et al. (2017) report difficulties with changing people's mindsets and beliefs. At the same time the literature review

demonstrates that digital leaders with a vision supported by empowered, knowledgeable and collaborative employees are critical to DT success. (Cichosz et al., 2020). During the times of digital transformation, as a change management strategy, it is very important for employees to be heard and respected, especially the employees who have been within the organization for a long time or have held an influential position at some point in the past (and believe they still do), otherwise they will assume that an important perspective has been missed and the answer can't be right if they have not been consulted (HBR, 2017), which consequently will turn them into the 'antagonists' of the organization's best intended transformation initiative. Kane et al. (2018) point out "competency traps" with employees being prisoners of their past successes.

According to HBR, 2022, The 27% of leaders are more likely to indicate that their organization has a much higher organizational commitment to cultural change compared to 14% of the remaining sample (Fig. 2).

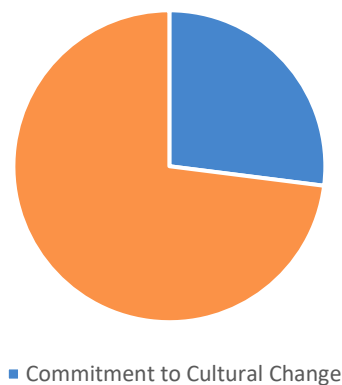


Fig. 3. Leaders committed to cultural change (HBR, 2022).

Different research across various industries highlights the change management as substantial factor in digitalization journey. Just to mention some, the examples from public, banking, logistics and production industries have been cited in here. In public institutions, organizational change is an imperative for appreciating the benefits of digitalization, a field that is currently expanding and evolving, and is prompting professionals and researchers to analyze and study the digitalization (Khalil Ibrahim, A., & Benabdelhadi, A., 2022). Decision-makers in banking sector are generally aware of digital change in banking and particularly aware of the change issue in digitalisation projects and that's why banks are even hiring a board member specifically focused on digitalisation, a newly emerging position, known as Chief Digital Officer (Cichosz et al., 2020). In the manufacturing sector, the research declared need for better education and skill training (individual) to improve the digital change (Vogelsang et al., 2019).

In May 2017, Health Metrics Advisory Team cited Sue Jauncey's, the founder and director of the organizational

culture firm Pulse Australasia (who are engaged to manage the UK's NHS change management strategy), statement: "In the majority of organizations, responsibility for cultural issues is all too often hived off to the HR department. However, there is now a growing recognition by corporate leaders and commentators that an organization's culture is the key responsibility of the board, CEO and Executive teams." And this statement coming from practice and few years ago is sustained by the study of Sariwulan et al. (2021), who present in their study that transformational leadership has the most significant influence on employee talent management rather than organizational culture and work division. The impact of leadership is related to the visionary view of achieving better results in the future, acting as agents of change, being transparent and democratic, giving trust to employees, and developing togetherness to achieve company progress. Companies can become industry leaders just by aligning people to achieve the CEO's vision. For this to work out, the business most important asset – people, need to align to the business strategy, by elevating their morale to encourage innovation and strengthen governance. On the other hand, employee talent management impacts job satisfaction, performance, and commitment to work sustainability (Pauli & Poczowski, 2019).

These statistics emphasize quite clearly the fact of how important it is to succeed with the digital transformation initiatives and how grounded is the need for achieving resilient and sustainable transformations to achieve strong and distinguished competitive advantage in business.

(HAT, 2017) highlights some of the key issues that contribute to failures when implementing change:

1. Poor communication both initially and on a regular basis
2. Employees who do not fully understand the reason for change
3. Insufficient training for employees preceding proposed changes
4. Failure to identify potential risk related to change
5. Failure to understand the effect and impact of changes
6. Flawed strategy to implement change
7. Failure to identify resources necessary to support change
8. Unrealistic timelines leading employees to feeling burdened and rushed.

There is another very interesting concept on Change Management found in the literature, and this is one of the cornerstone models for understanding organizational change which was developed by Kurt Lewin back in the 1940s, and still holds true today (Schein, 2004). His model is known as Unfreeze – Change – Refreeze, which refers to the three-stage process of change that he describes. Lewin, a physicist as well as a social scientist, explained organizational change using the analogy of changing the shape of a block of ice. If you have a large cube of ice but realize that what you want is a cone of ice, what do you do? First you must melt the ice to make it

amenable to change (unfreeze). Then you must mold the iced water into the shape you want (change). Finally, you must solidify the new shape (refreeze).

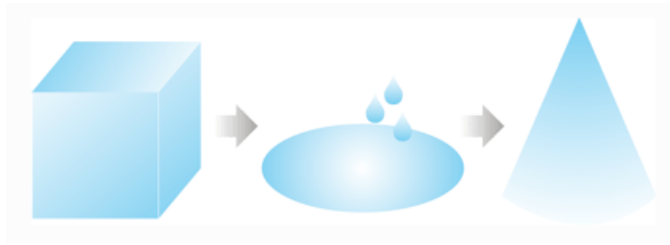


Fig. 4. Lewin's Ice Cube example for Change Management

The literature review on change management through time brings us to a conclusion that, the well-established, classic change management levers used in information system implementation projects should be adopted to the context of digital transformation as well. The set of levers for Change Management – comprised of *leadership, teams, culture, change story, communication, training, role transitioning and learning organization* – remain the same but should be tuned to the requirements of digitalization (Siemens, 2019).

4. DISCUSSION

Study shows that digital transformation is expanding as a trend at an enormous pace, but the risk of failure remains still high. These failures largely derive from the unique challenges associated with digital transformation, including shifting from the current culture based on papers to a more digital and entrepreneurial one, dealing with a lack of digital talent, working in cross-functional teams, meeting short deadlines due to higher customer expectations, and evolving target states instead of fixed goals.

Not enough focus is given to the 'transformation' part of digital transformation, and the transformation part has always been about people. This has been the blind spot for so many digital transformation efforts, it's what a lot of companies are missing. Digital Transformation initiatives are complex projects because they introduce new processes, new organizational culture, and new technologies, so the employee's resistance to adapt is understandable; and deploying these projects with success is not an easy task nor a straight-forward process. Even the definition of success per se is ambiguous. As resulted from one of my earlier studies (Pacolli, 2018), even if digital transformation project completes with success, i.e., on time, on budget and with all the requirements met, still if the deployed solution is not fully used and by all the intended users/employees, than the whole initiative could turn into a failure.

Therefore, enlightening the importance of sustainability and resilience in digital transformation is very important. And literature analysis shows that if we give the change management correct attention, then the journey towards a sustainable digital transformation is guaranteed.

What most business leaders underestimate is that digital transformation is not just about technology: above all, it is about people and that's what makes it more complex, thus requiring a new approach for managing the change and develop a framework for sustainable digital transformation. The study results suggest three key imperatives for successful and sustainable digital transformations. These imperatives have formed our cohesive Change Management Framework: Leadership, Communication, Inclusion.



Fig. 5. Change Management Framework

5. CONCLUSION

The Covid-19 pandemic has been a catalyst for many organizations in pursuing digital transformation initiative, leading companies to accelerate technology initiatives, design dramatic changes to processes, and reexamine culture for change management and the role it plays in today's new business world. Despite the change culture increasing in importance, organizations appear to be having trouble improving it: just 31% rate their organization's performance in modernizing workplace cultures in 2021 as being effective (Harvard Business Review Report, 2021).

Digital transformation tackles mostly the employees as it transforms their usual way of working, and it is in human nature not show acceptance at the beginning of a change and often reject or resist the new things at first. Employees have to be involved in the change process and learn how to deal with digitalisation and corresponding innovations. (Diener et al., 2021).

Organizational changes very often tend to grow some strange, enemy-like, emotions within employees towards the organization. So, if the organizations follow this framework, their employees will be their real team and when you have a team you chase the same objectives thus the risk of DT initiatives failure will be much lower, and the solution will sustain long after the project is delivered. The convergence of digitization and sustainability is providing new tools that enable enterprises to make a lasting impact on the planet's natural capital (George, G. and Schillebeekx, S.J.D., 2022).

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